From Average and Difference to the Wedge

What every VSA binding keeps, what it drops, and the object that keeps both

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Where This Bridge Goes

Today's binding operations, MAP's coordinatewise product and HRR's recipe of shifts, share one property: both are blind to order. dog bound to man is the same vector as man bound to dog, and the roles or the permutation had to be added to say who bit whom.

Earlier in the term, the average/difference sessions gave us a tool for saying exactly what an order-blind operation drops: split any two-slot object into a **symmetric half** and an **antisymmetric half**.

I	The split, recalled	numbers, functions, the attention head, a bound pair
II	What today's bindings keep	HRR and MAP keep S and drop A, exactly; the impasse
III	The wedge	the same split applied to a *product*; the dropped half, kept
IV	Binding with the wedge	order in one bit; exact unbinding; what it does not give

Friday's session says where such a product comes from: one declared rule.

One Move, Recalled: Average and Half-Difference

The average/difference sessions turned on one move. Any two numbers can be traded for their average and their half-difference, and recovered from them:

$$a = \underbrace{\tfrac{a+b}{2}}_{\text{average}} + \underbrace{\tfrac{a-b}{2}}_{\text{half-difference}}, \qquad b = \tfrac{a+b}{2} - \tfrac{a-b}{2}$$

Nothing is lost: two numbers in, two numbers out. What changes is what the two numbers *say*:

	under the swap $a \leftrightarrow b$	**answers**
the average $\frac{a+b}{2}$	unchanged	how much, in common
the half-difference $\frac{a-b}{2}$	changes sign	which is bigger, and by how much

The same move splits a function into even and odd parts, $f(x) = \frac{1}{2}[f(x) + f(-x)] + \frac{1}{2}[f(x) - f(-x)]$, with the swap $x \leftrightarrow -x$: $e^x = \cosh x + \sinh x$, and at $x = 1$ that is $2.718 = 1.543 + 1.175$.

Everything today is this move, applied to a different pair. The pair for a binding is "$\mathbf{u}$ first" and "$\mathbf{v}$ first."

The Move on Matrices: The Head, Split

For a square matrix, "swap" is the transpose, and the move gives

$$G = \underbrace{\tfrac{1}{2}(G + G^\top)}_{S,\ \text{symmetric}} + \underbrace{\tfrac{1}{2}(G - G^\top)}_{A,\ \text{antisymmetric}}$$

Session 11's attention head, scoring $\mathbf{x}_1 = (3, 1)$ against $\mathbf{x}_2 = (1, 3)$:

$G = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right) + \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$	$1 \to 2$	$2 \to 1$	
$\mathbf{x}^\top S\,\mathbf{y}$	16	16	how alike
$\mathbf{x}^\top A\,\mathbf{y}$	$+8$	-8	which way
$\mathbf{x}^\top G\,\mathbf{y}$	24	8	the score

S answers "how alike"; A answers "which way." And the two scores split by the move on numbers: 24 and 8 have average 16 and half-difference 8. Order lives in A, and only in A.

Why Order Lives Only in A

Two facts about any antisymmetric matrix A, each one line:

Swapping the arguments flips the sign

$$\mathbf{y}^\top A\,\mathbf{x} = (\mathbf{y}^\top A\,\mathbf{x})^\top = \mathbf{x}^\top A^\top \mathbf{y} = -\,\mathbf{x}^\top A\,\mathbf{y}$$

A number equals its own transpose; $A^\top = -A$ does the rest.

The same argument twice gives 0

$$\mathbf{x}^\top A\,\mathbf{x} = -\,\mathbf{x}^\top A\,\mathbf{x} \quad\Longrightarrow\quad \mathbf{x}^\top A\,\mathbf{x} = 0$$

A quantity equal to its own negative is zero.

The head: $\mathbf{x}_1^\top A\,\mathbf{x}_2 = 8$, $\mathbf{x}_2^\top A\,\mathbf{x}_1 = -8$, and $\mathbf{x}_1^\top A\,\mathbf{x}_1 = 0$.

A linguistic reading

If a relation were stored as A alone, *Mary likes Bob* and *Bob likes Mary* would score $+8$ and -8, and *Mary likes herself* would score 0 whoever Mary is. A is a pure "which way" detector. S is a pure "how alike" detector: it cannot tell the two orders apart. A relation that must do both needs both.

The Same Split on a Bound Pair

index	0	1	2	3
$\mathbf{u}$	1	2	0	1
$\mathbf{v}$	2	0	1	1

Session 11c started binding from the outer product, the 4×4 table of all products $u_i v_j$, because it tells $\mathbf{u}\mathbf{v}^\top$ from $\mathbf{v}\mathbf{u}^\top$. Split it:

$$S = \tfrac{1}{2}(M + M^\top) = \begin{pmatrix} 2 & 2 & \frac{1}{2} & \frac{3}{2} \\ 2 & 0 & 1 & 1 \\ \frac{1}{2} & 1 & 0 & \frac{1}{2} \\ \frac{3}{2} & 1 & \frac{1}{2} & 1 \end{pmatrix} \qquad A = \tfrac{1}{2}(M - M^\top) = \begin{pmatrix} 0 & -2 & \frac{1}{2} & -\frac{1}{2} \\ 2 & 0 & 1 & 1 \\ -\frac{1}{2} & -1 & 0 & -\frac{1}{2} \\ \frac{1}{2} & -1 & \frac{1}{2} & 0 \end{pmatrix}$$

	free numbers	**what it records**
S	10 (the diagonal and one triangle)	how much of each product; its diagonal sums to $\mathbf{u} \cdot \mathbf{v} = 3$
A	6 (one triangle; the diagonal is 0)	which vector came first: swap $\mathbf{u}, \mathbf{v}$ and A flips sign

Sixteen numbers in all, ten plus six. Any binding that returns four numbers must give most of this up. The question is which part.

One Move, Four Places

object	the swap	symmetric half	antisymmetric half	where
two numbers a, b	$a \leftrightarrow b$	average	half-difference	average/difference
a function f	$x \leftrightarrow -x$	even part	odd part	average/difference
a matrix G	transpose	$S = \frac{1}{2}(G + G^\top)$	$A = \frac{1}{2}(G - G^\top)$	11, the head
a bound pair $\mathbf{u}\mathbf{v}^\top$	$\mathbf{u} \leftrightarrow \mathbf{v}$ (transpose)	$\frac{1}{2}(\mathbf{u}\mathbf{v}^\top + \mathbf{v}\mathbf{u}^\top)$	$\frac{1}{2}(\mathbf{u}\mathbf{v}^\top - \mathbf{v}\mathbf{u}^\top)$	11c
a product $\mathbf{uv}$	$\mathbf{uv} \leftrightarrow \mathbf{vu}$	the inner product	the wedge	Part III, today

In every row the two halves together hold everything, the symmetric half is blind to the swap, and the antisymmetric half is where order lives.

Row four is row three on one particular matrix: transposing the outer product swaps the vectors, $(\mathbf{u}\mathbf{v}^\top)^\top = \mathbf{v}\mathbf{u}^\top$, so for a bound pair "transpose" and "$\mathbf{u} \leftrightarrow \mathbf{v}$" are the same move. That makes the last row a step, not a leap: $\mathbf{uv} \leftrightarrow \mathbf{vu}$ is the same swap on a different product of the same two vectors. The move does not care whether the object is a matrix or a product.

First, Part II: which half do today's bindings keep?

HRR Keeps S and Drops A, Exactly

The fold that makes circular convolution adds the cells of $\mathbf{u}\mathbf{v}^\top$ along anti-diagonals. Folding is linear, so fold the halves separately (Session 11c):

what is folded	**result**	
$M = \mathbf{u}\mathbf{v}^\top$	$(4, 5, 2, 5) = \mathbf{u} \circledast \mathbf{v}$	the binding
the symmetric half S	$(4, 5, 2, 5)$	all of it
the antisymmetric half A	$(0, 0, 0, 0)$	none of it

Transposing moves cell (i, j) to (j, i), in the same anti-diagonal, so each group sum of A equals its own negative: exactly 0.

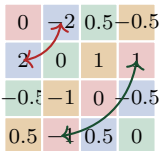
Circular convolution is the symmetric half of the outer product, folded. It is blind to order not by accident but because the order-carrying half of its input is exactly what the fold removes. An operation built from S alone could not have been anything but commutative.

The next slide shows the mechanism in a picture; the one after does the same accounting for MAP.

Why the Fold Kills A: The Picture

The fold adds cells of the same color, where the color of cell (i, j) is $(i + j) \bmod 4$. Transposing sends (i, j) to (j, i), and $j + i = i + j$: the transpose moves cells only within a color.

$A =$



cells $(0, 1)$ and $(1, 0)$: both color 1, values -2 and 2
cells $(1, 3)$ and $(3, 1)$: both color 0, values 1 and -1
every off-diagonal cell has its negative in its own color,
and the diagonal is 0: each color sums to 0

The one-line proof

Let a_k be the color-k sum of A. Transposing permutes color k onto itself, so $a_k(A^\top) = a_k(A)$. But $A^\top = -A$, so $a_k(A^\top) = -a_k(A)$. Hence $a_k = -a_k = 0$. No property of $\mathbf{u}$ or $\mathbf{v}$ entered, so this holds for every pair of vectors in every dimension.

MAP Keeps Even Less of S, and Also None of A

MAP binds ± 1 vectors coordinate by coordinate. With dog $= (+,-,+,+)$ and man $= (-,-,+,-)$:

$$\text{dog} \otimes \text{man} = (-,+,+,-) = \text{man} \otimes \text{dog}$$

In the outer-product table $\text{dog}\,\text{man}^\top$, these four numbers are the **diagonal**, the cells (i,i). The diagonal belongs entirely to S, since A's diagonal is 0: MAP never looks at a cross term at all.

binding	**keeps of S**	**keeps of A**	**so**
HRR, $\mathbf{u} \circledast \mathbf{v}$	the four anti-diagonal sums	nothing	commutes
MAP, $\mathbf{u} \otimes \mathbf{v}$	the four diagonal cells	nothing	commutes

Both of today's bindings keep part of the symmetric half and none of the antisymmetric half. That is why both commute, and why order had to be put back from outside: by role vectors ($\mathbf{agt} \otimes \text{dog} + \mathbf{pat} \otimes \text{man}$) or by the permutation ρ that marks position.

The roles are the patch. They work; the question here is whether there is an operation that does not need it.

The Impasse, Counted

A natural thought: if the fold destroys A, carry A separately. Count what that costs:

n	S **holds**	A **holds**	**the fold keeps**
4	10	6	4
300	45,150	44,850	300
1000	500,500	499,500	1000

A is just as quadratic as S: $\binom{n}{2}$ numbers. Carrying it puts us back at n^2, the outer product we were trying to escape.

So the VSA trade is a real one: pay n and supply order by convention, or pay n^2 and keep it. Today's learning segment showed the first option's other cost: everything stored shares one space, so facts blur into one another.

Part III is a third option. It does not make A smaller. It changes *what kind of object* holds A, so that the result can be multiplied again.

Apply the Split to the Product Itself

So far the split was applied to a *matrix*. Apply it instead to a *product*. Suppose vectors can be multiplied, $\mathbf{uv}$, in a way that need not commute. Then half-sum and half-difference give, for any such product,

$$\mathbf{uv} \;=\; \underbrace{\tfrac{1}{2}(\mathbf{uv}+\mathbf{vu})}_{\text{symmetric half}} \;+\; \underbrace{\tfrac{1}{2}(\mathbf{uv}-\mathbf{vu})}_{\text{antisymmetric half}}$$

This is the same identity as $a = \frac{a+b}{2} + \frac{a-b}{2}$, with $a = \mathbf{uv}$ and $b = \mathbf{vu}$. Everything true of S and A is true of these halves: swap $\mathbf{u}$ and $\mathbf{v}$, and the first stays while the second flips sign.

half	**name**	**what it turns out to be**
$\frac{1}{2}(\mathbf{uv}+\mathbf{vu})$	the inner product, $\mathbf{u}\cdot\mathbf{v}$	a number: how alike
$\frac{1}{2}(\mathbf{uv}-\mathbf{vu})$	the **wedge**, $\mathbf{u}\wedge\mathbf{v}$	a new kind of object: which way, in which plane

The wedge is defined as the antisymmetric half of a product. Nothing has been assumed about the product except that it distributes over addition. The next slide uses only that.

The Wedge in Two Dimensions, Computed

Take the plane, $\mathbf{u} = u_1\mathbf{e}_1 + u_2\mathbf{e}_2$ and $\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2$, and multiply out, keeping every product of basis vectors in the order it arrives:

$$\mathbf{uv} = u_1v_1\,\mathbf{e}_1\mathbf{e}_1 + u_1v_2\,\mathbf{e}_1\mathbf{e}_2 + u_2v_1\,\mathbf{e}_2\mathbf{e}_1 + u_2v_2\,\mathbf{e}_2\mathbf{e}_2$$
$$\mathbf{vu} = v_1u_1\,\mathbf{e}_1\mathbf{e}_1 + v_1u_2\,\mathbf{e}_1\mathbf{e}_2 + v_2u_1\,\mathbf{e}_2\mathbf{e}_1 + v_2u_2\,\mathbf{e}_2\mathbf{e}_2$$

Subtract. The $\mathbf{e}_1\mathbf{e}_1$ and $\mathbf{e}_2\mathbf{e}_2$ terms have the same coefficient in both lines and cancel. The cross terms do not:

$$\tfrac{1}{2}(\mathbf{uv} - \mathbf{vu}) \;=\; (u_1v_2 - u_2v_1)\cdot\tfrac{1}{2}(\mathbf{e}_1\mathbf{e}_2 - \mathbf{e}_2\mathbf{e}_1) \;=\; (u_1v_2 - u_2v_1)\,\mathbf{e}_1\wedge\mathbf{e}_2$$

- The coefficient $u_1v_2 - u_2v_1$ is the 2×2 determinant, and it is $2A_{12}$ for $A = \frac{1}{2}(\mathbf{uv}^\top - \mathbf{vu}^\top)$.
- The object $\mathbf{e}_1\wedge\mathbf{e}_2$ is the antisymmetric half of $\mathbf{e}_1\mathbf{e}_2$ itself: the unit plane.
- Nothing was assumed about $\mathbf{e}_1\mathbf{e}_1$, or about how $\mathbf{e}_1\mathbf{e}_2$ relates to $\mathbf{e}_2\mathbf{e}_1$. Those are what Friday's declaration settles. The wedge did not need them.

The Wedge Is A, Kept

index	0	1	2	3
$\mathbf{u}$	1	2	0	1
$\mathbf{v}$	2	0	1	1

The last slide's computation runs the same way in any dimension: one coefficient per *pair* of directions,

$$\mathbf{u} \wedge \mathbf{v} \;=\; \sum_{i<j} (u_i v_j - u_j v_i)\, \mathbf{e}_i \wedge \mathbf{e}_j$$

For the hand vectors, in four dimensions, there are six pairs, so six coordinate planes:

plane (i, j)	$(0,1)$	$(0,2)$	$(0,3)$	$(1,2)$	$(1,3)$	$(2,3)$
$u_i v_j - u_j v_i$	-4	1	-1	2	2	-1
$2A_{ij}$, from the split of $\mathbf{u}\mathbf{v}^\top$ in Part I	-4	1	-1	2	2	-1

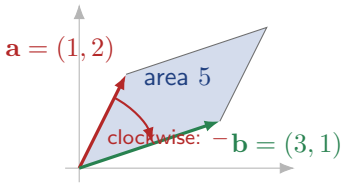
The six coordinates of the wedge are the six free numbers of A, doubled: the half the fold annihilated, entry for entry. And the inner product is $\mathbf{u} \cdot \mathbf{v} = 3$, the trace of S.

$\mathbf{u}\mathbf{v} = 3 + \mathbf{u} \wedge \mathbf{v}$ and $\mathbf{v}\mathbf{u} = 3 - \mathbf{u} \wedge \mathbf{v}$. One number from S, all of A. The fold kept four numbers from S and nothing from A.

What the Wedge Looks Like

In the plane there is only one pair of directions, so the wedge has one coordinate. For $\mathbf{a} = (1, 2)$ and $\mathbf{b} = (3, 1)$:

$$\mathbf{a} \wedge \mathbf{b} = (1 \cdot 1 - 2 \cdot 3)\, \mathbf{e}_1 \wedge \mathbf{e}_2 = -5\, \mathbf{e}_1 \wedge \mathbf{e}_2$$



- $|-5|$ is the **area** of the parallelogram the two vectors span.
- The sign is the **direction of turn** from $\mathbf{a}$ to $\mathbf{b}$: here clockwise, so negative. Going from $\mathbf{b}$ to $\mathbf{a}$ gives $+5$.
- $\mathbf{e}_1 \wedge \mathbf{e}_2$ names the **plane**. In four dimensions there are six such planes, which is why the wedge above had six coordinates.

Compare the dot product of the same two vectors, $\mathbf{a} \cdot \mathbf{b} = 3 + 2 = 5$: a single number, the same in either order. The pair $(5,\ -5\, \mathbf{e}_1 \wedge \mathbf{e}_2)$ is $\mathbf{ab}$, and $(5,\ +5\, \mathbf{e}_1 \wedge \mathbf{e}_2)$ is $\mathbf{ba}$.

Three Properties, Each From the Formula

With $\mathbf{a} = (1, 2)$ and $\mathbf{b} = (3, 1)$, so that $\mathbf{a} \wedge \mathbf{b} = -5\,\mathbf{e}_1 \wedge \mathbf{e}_2$:

property	**from the formula**	**checked**	**meaning**
$\mathbf{b} \wedge \mathbf{a} = -\,\mathbf{a} \wedge \mathbf{b}$	$b_1a_2 - b_2a_1 = -(a_1b_2 - a_2b_1)$	$3 \cdot 2 - 1 \cdot 1 = +5$	the order bit lives in the sign
$\mathbf{a} \wedge \mathbf{a} = 0$	$a_1a_2 - a_2a_1 = 0$	$1 \cdot 2 - 2 \cdot 1 = 0$	no plane from one vector; A's zero diagonal
$\mathbf{a} \wedge (\mathbf{b} + 2\mathbf{a}) = \mathbf{a} \wedge \mathbf{b}$	$\mathbf{a} \wedge \mathbf{b} + 2\,\mathbf{a} \wedge \mathbf{a}$	$\mathbf{b} + 2\mathbf{a} = (5, 5)$, and $1 \cdot 5 - 2 \cdot 5 = -5$	sliding $\mathbf{b}$ along $\mathbf{a}$ changes nothing

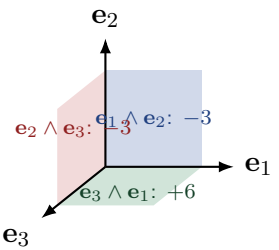
The third property says the wedge **forgets on purpose**: many pairs give the same plane, orientation and area. Compare the dot product, which forgets the other things: $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a} = 5$, and the order is gone.

Between them the two halves keep what matters about the pair: how alike (5), and which way, in which plane, with what area $(-5\,\mathbf{e}_1 \wedge \mathbf{e}_2)$.

In Three Dimensions: One Number per Plane

Three directions, three coordinate planes, three coordinates. For $\mathbf{a} = (1, 2, 3)$ and $\mathbf{b} = (4, 5, 6)$:

plane	coefficient	
$\mathbf{e}_1 \wedge \mathbf{e}_2$	$a_1b_2 - a_2b_1 = 5 - 8$	-3
$\mathbf{e}_2 \wedge \mathbf{e}_3$	$a_2b_3 - a_3b_2 = 12 - 15$	-3
$\mathbf{e}_3 \wedge \mathbf{e}_1$	$a_3b_1 - a_1b_3 = 12 - 6$	$+6$



three planes, three coordinates

Each number is how much of the parallelogram's oriented area projects onto that plane; together they name a tilted plane with an area and a turn. Readers who know the cross product will recognize $(-3, 6, -3)$ rearranged; 13c says why that works only in 3D.

In n dimensions there are $\binom{n}{2}$ coordinate planes: 1, 3, 6, and at $n = 300$, 44,850. The same count as the free entries of A, as it must be.

What Each Construction Keeps of the Pair

construction	**of S**	**of A**	$n = 4$	$n = 300$	**order?**
outer product $\mathbf{u}\mathbf{v}^\top$	all 10	all 6	16	90,000	kept
HRR, $\mathbf{u} \circledast \mathbf{v}$	four anti-diagonal sums	nothing	4	300	lost
MAP, $\mathbf{u} \otimes \mathbf{v}$	the four diagonal cells	nothing	4	300	lost
the product $\mathbf{u}\mathbf{v} = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \wedge \mathbf{v}$	one number: the trace	all 6	7	44,851	kept

Read the last row against the impasse. It does not make A cheap: 44,850 numbers at $n = 300$, as counted. What it changes is two other things:

- **Which part of S survives.** The fold kept four sums that mean nothing in particular; the product keeps the one number that means "how alike," the inner product.
- **What kind of object the result is.** A matrix of A's entries is a dead end: you cannot multiply it by another vector and stay in the same algebra. A wedge can be multiplied again, which is what binding a bound pair requires. Friday and 13c show how.

Binding With the Wedge: A Preview

Put today's two sentences through each binding:

	dog bites man	*man bites dog*	
MAP, bare	$\text{dog} \otimes \text{man}$	$\text{man} \otimes \text{dog}$	the same vector
MAP, with roles	$\mathbf{agt} \otimes \text{dog} + \mathbf{pat} \otimes \text{man}$	$\mathbf{agt} \otimes \text{man} + \mathbf{pat} \otimes \text{dog}$	different: the roles were supplied
wedge, bare	$\text{dog} \wedge \text{man}$	$\text{man} \wedge \text{dog} = -\,\text{dog} \wedge \text{man}$	different, by the sign

The wedge records who came first without a role vector, in one bit. Roles are still wanted, for naming the positions and for three or more arguments; the point is where the order information lives.

Three things change, and Session 13c works them out:

- The bound pair has a **new type**: it is a plane, not a vector, so nothing can mistake it for a filler.
- Unbinding with an orthogonal role is **exact**, with no cleanup, where today's unbinding returned the filler plus noise.
- The price is the one from the impasse: a pair costs $\binom{n}{2}$ coordinates, not n.

Unbinding With the Wedge: A Worked Preview

Give roles their own directions, orthogonal to every filler: $\mathbf{r}_{\text{ag}} = \mathbf{e}_1$, $\mathbf{r}_{\text{th}} = \mathbf{e}_2$; fillers $\text{john} = \mathbf{e}_3$ and $\text{mary} = 0.6\,\mathbf{e}_3 + 0.8\,\mathbf{e}_4$. Bind each role to its filler with the wedge and add:

$$\mathbf{E} = \mathbf{r}_{\text{ag}} \wedge \text{john} + \mathbf{r}_{\text{th}} \wedge \text{mary} = \mathbf{e}_1 \wedge \mathbf{e}_3 + 0.6\,\mathbf{e}_2 \wedge \mathbf{e}_3 + 0.8\,\mathbf{e}_2 \wedge \mathbf{e}_4$$

Who is the agent? Look at the planes that contain $\mathbf{e}_1$. There is one, $\mathbf{e}_1 \wedge \mathbf{e}_3$, with coefficient 1, so the partner of $\mathbf{e}_1$ is $\mathbf{e}_3 = \text{john}$. **Who is the theme?** The planes containing $\mathbf{e}_2$ are $\mathbf{e}_2 \wedge \mathbf{e}_3$ and $\mathbf{e}_2 \wedge \mathbf{e}_4$, with coefficients 0.6 and 0.8: the partner is $0.6\,\mathbf{e}_3 + 0.8\,\mathbf{e}_4 = \text{mary}$.

	HRR / MAP (Session 12)	**wedge**
query returns	the filler plus noise from the other fields	the filler, exactly
cleanup needed	yes	no
why	every field shares one space	each role has planes of its own

"Read off the partner" is an operation with a name, **contraction**, and Session 13c defines it. The condition for exactness is the one used here: roles orthogonal to fillers, and to each other.

What the Wedge Does Not Give for Free

- **The order bit is one bit.** $\text{dog} \wedge \text{man} = -\,\text{man} \wedge \text{dog}$ says the two orders differ. Which sign means "agent first" is a convention, like the right-hand rule. For two arguments the sign is enough; for naming positions, and for three or more arguments, roles are still wanted.
- **The cost stands.** A bound pair has $\binom{n}{2}$ coordinates, not n. A bound triple $\mathbf{a} \wedge \mathbf{b} \wedge \mathbf{c}$ has $\binom{n}{3}$. The wedge keeps order by paying for it, which is what the impasse said any order-keeping scheme must do.
- **A pair is a new type.** It cannot be mistaken for a filler, which is a gain for knowing what you have and a loss for superposing everything in one space, which is what made HRR's cleanup memory work.
- **Exactness needs orthogonality.** With random high-dimensional roles and fillers, orthogonality is approximate and some noise returns. How much, compared with HRR at the same dimension, is an empirical question, and a possible project.

So the wedge is not a free upgrade of today's bindings. It is the other side of the trade: order and exact recovery, paid for in size and in the loss of the single space.

Where the Product Comes From: Friday

These slides took the product $\mathbf{uv}$ for granted and showed one thing about it: whatever it is, half-sum and half-difference split it into a symmetric part and the wedge.

Friday's question

What rule defines a product of vectors? A single declaration, **what each vector squared is**: $\mathbf{vv} = Q(\mathbf{v})$, a number. From that rule alone the symmetric half is forced, by the half-sum applied to $(\mathbf{u}+\mathbf{v})^2$, and the antisymmetric half comes free: the rule never mentions it.

session	question	what it adds
13a, Friday	what is declared?	the squaring rule, the signature, the product with both halves
13b	what does the rule do?	the rule as an interpreter: turns, boosts, undo
13c	what is built on it?	the wedge as a constructor, the dual as a cast, wedge binding

The thread from the average/difference sessions runs straight through: split, keep both halves, ask what each half answers. Today's bindings answered "how alike." The wedge answers "which way."

Summary

1. One move, average and half-difference, splits any two-slot object into a symmetric half (blind to the swap) and an antisymmetric half (flips under it). Order lives only in the antisymmetric half.
2. A bound pair $\mathbf{u}\mathbf{v}^\top$ splits into S (10 numbers at $n = 4$) and A (6). HRR's fold keeps four sums of S and annihilates A exactly; MAP keeps the diagonal of S and never sees A. That is why both commute and both need roles.
3. Carrying A separately costs $\binom{n}{2}$: the impasse.
4. Apply the same move to a product $\mathbf{u}\mathbf{v}$: the symmetric half is the inner product, and the antisymmetric half is the **wedge**. Expanding in a basis needs no assumption about the product; the wedge's coordinates are the entries of A, doubled.
5. The wedge is an oriented plane with an area. It is anticommutative, vanishes on a repeated vector, and forgets on purpose.
6. Bound with the wedge, a role–filler pair keeps its order in the sign and can be unbound exactly, at the price of size and of the single space.